

Worksheet - Lecture 8

Characterizing Solution Sets

1. For each of the following systems, determine whether they have a unique solution, no solution, or infinitely many solutions. If a system has a unique solution, provide that solution. If a system has infinitely many solutions, characterize the solution set as was done in the lecture.

a.
$$\begin{cases} x + 2y + z = 2 \\ 2x + 4y = 2 \\ 3x + 6y + z = 4 \end{cases}$$

$$\begin{pmatrix} 1 & 2 & 1 & | & 2 \\ 2 & 4 & 0 & | & 2 \\ 3 & 6 & 1 & | & 4 \end{pmatrix} \xrightarrow{\substack{R_2' = R_2 - 2R_1 \\ R_3' = R_3 - 3R_1}} \begin{pmatrix} 1 & 2 & 1 & | & 2 \\ 0 & 0 & -2 & | & -2 \\ 0 & 0 & -2 & | & -2 \end{pmatrix}$$

$$\xrightarrow{\substack{R_3' = R_3 - R_2 \\ R_2' = -\frac{1}{2}R_2}} \begin{pmatrix} 1 & 2 & 1 & | & 2 \\ 0 & 0 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{pmatrix} \xrightarrow{R_1' = R_1 - R_2} \begin{pmatrix} 1 & 2 & 0 & | & 1 \\ 0 & 0 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$x_1 + 2x_2 = 1$
 $x_3 = 1$
 $x_1 = 1 - 2x_2$
 $x_2 \equiv \text{free}$
 $x_3 = 1$
 let $x_2 = s$
 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 - 2s \\ s \\ 1 \end{pmatrix}$

b.
$$\begin{cases} 2x_1 + 2x_2 + 6x_3 = 4 \\ 2x_1 + x_2 + 7x_3 = 6 \\ -2x_1 - 6x_2 - 7x_3 = -1 \end{cases}$$

$$\begin{pmatrix} 2 & 2 & 6 & | & 4 \\ 2 & 1 & 7 & | & 6 \\ -2 & -6 & -7 & | & -1 \end{pmatrix}$$

unique solution: $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 6 \\ 1 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

c.
$$\begin{cases} -h + l + w = 1 \\ -h - l + w = 2 \\ h - l + w = 3 \\ h + l + w = 4 \end{cases}$$

$$\begin{pmatrix} -1 & 1 & 1 & | & 1 \\ -1 & -1 & 1 & | & 2 \\ 1 & -1 & 1 & | & 3 \\ 1 & 1 & 1 & | & 4 \end{pmatrix} \xrightarrow{\substack{R_2' = R_2 - R_1 \\ R_3' = R_3 + R_1 \\ R_4' = R_4 + R_1}} \begin{pmatrix} -1 & 1 & 1 & | & 1 \\ 0 & -2 & 0 & | & 1 \\ 0 & 0 & 2 & | & 4 \\ 0 & 2 & 2 & | & 5 \end{pmatrix} \xrightarrow{R_4' = R_4 + R_2} \begin{pmatrix} -1 & 1 & 1 & | & 1 \\ 0 & -2 & 0 & | & 1 \\ 0 & 0 & 2 & | & 4 \\ 0 & 0 & 2 & | & 6 \end{pmatrix}$$

$$\xrightarrow{R_4' = R_4 - R_3} \begin{pmatrix} -1 & 1 & 1 & | & 1 \\ 0 & -2 & 0 & | & 1 \\ 0 & 0 & 2 & | & 4 \\ 0 & 0 & 0 & | & 2 \end{pmatrix}$$

INCONSISTENT

d.
$$\begin{cases} -h + l + w = 1 \\ -h - l + w = 2 \\ h - l + w = 3 \\ h + l + w = 2 \end{cases}$$

Same steps as above, only last row changes.

$$\begin{pmatrix} -1 & 1 & 1 & | & 1 \\ 0 & -2 & 0 & | & 1 \\ 0 & 0 & 2 & | & 4 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -1/2 \\ 2 \end{pmatrix}$ unique solution

e.
$$\begin{cases} x_1 + 2x_2 + 2x_3 + 3x_4 = 0 \\ 2x_1 + 4x_2 + x_3 + 3x_4 = 0 \\ 3x_1 + 6x_2 + x_3 + 4x_4 = 0 \end{cases}$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & | & 0 \\ 2 & 4 & 1 & 3 & | & 0 \\ 3 & 6 & 1 & 4 & | & 0 \end{pmatrix} \xrightarrow{\substack{R_2' = R_2 - 2R_1 \\ R_3' = R_3 - 3R_1}} \begin{pmatrix} 1 & 2 & 2 & 3 & | & 0 \\ 0 & 0 & -3 & -3 & | & 0 \\ 0 & 0 & -5 & -5 & | & 0 \end{pmatrix}$$

$$\xrightarrow{\substack{R_2' = -\frac{1}{3}R_2 \\ R_3' = -\frac{1}{5}R_3}} \begin{pmatrix} 1 & 2 & 2 & 3 & | & 0 \\ 0 & 0 & 1 & 1 & | & 0 \\ 0 & 0 & 1 & 1 & | & 0 \end{pmatrix} \xrightarrow{R_3' = R_3 - R_2} \begin{pmatrix} 1 & 2 & 2 & 3 & | & 0 \\ 0 & 0 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$$\xrightarrow{R_1' = R_1 - 2R_2} \begin{pmatrix} 1 & 2 & 0 & 1 & | & 0 \\ 0 & 0 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$x_1 = -2x_2 - x_4$
 $x_2 = \text{free}$
 $x_3 = -x_4$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = s \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$